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RESEARCH ARTICLE

Enhancing Worker Safety at Heights: A Deep Learning Model for Detecting Helmets and Harnesses Using DETR Architecture

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ABSTRACT We propose an approach for monitoring work-at-heights activities by ensuring the presence of safety helmets and harnesses in construction environments using the DEtection TRansformer (DETR) model. The DETR-based model is trained and tested on a custom dataset to accurately detect these essential safety items. The model's performance was evaluated using standard metrics. For helmet detection, it achieved an average precision of 0.99, a recall of 0.92, an F1 score of 0.95, and a confidence rate exceeding 95%. Similarly, for harness detection, the model attained an average precision of 0.971, a recall of 0.91, an F1 score of 0.94, and a confidence rate above 93%. To evaluate its reliability, the model was tested under diverse conditions, including RGB, grayscale, blurred, and dusty images, as well as scenarios involving multiple helmets and harnesses. It was also assessed using drone-captured images to ensure robust performance across varying weather and lighting conditions. The results were then compared with a previously trained YOLOv3 model developed for the same detection task, as well as two newly trained models based on the YOLOv8 and Deformable DETR architectures. Our analysis demonstrates that the DETR model not only improves detection accuracy but also overcomes certain limitations inherent in other models. The comparative study highlights key architectural and performance differences, providing insights into the advantages of transformer-based models for object detection tasks in industrial safety applications.

INDEX TERMS Deep learning, detection transformer (DETR), construction safety, machine learning.

I. INTRODUCTION

The construction industry is one of the largest sectors globally, generating billions of dollars for national economy [1]. However, it faces many challenges in ensuring worker safety during project execution. According to the Bureau of Labor Statistics, construction remains the most hazardous industry, accounting for a substantial portion of workplace fatalities. In 2023 alone, 1 in 5 fatal workplace injuries occurred on construction sites. Additionally, out of 345 fall-related deaths reported that year, 96 were in the construction industry [2]. In response to these alarming statistics, many countries have implemented stringent regulations to enhance worker safety. One of the most effective measures for reducing fatalities is the enforcement of strict policies regarding Personal Protective Equipment (PPE) and Personal Fall Arrest Systems (PFAS). These safety equipment play a crucial role in mitigating risks and ensuring a safer work environment for construction workers engaged in work at height. A safety helmet is a type of protective headgear designed to safeguard the head from injuries caused by falling objects and impacts in various environments. Similarly, safety

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harness is a protective device worn by workers to secure them to an anchor point, preventing falls or minimizing injuries during high-risk activities or tasks performed at heights. Figure 1 illustrates a typical safety helmet and harness commonly used on construction sites.

Despite the implementation of strict policies requiring the use of protective gear, injuries and fatalities continue to occur, often due to worker incompetence. This arose a need for automated and technology-driven techniques to assess worker competency, rather than relying solely on traditional monitoring methods (e.g., safety inspectors). Since safety inspectors cannot be present in all locations simultaneously, automated safety inspection and monitoring tools should be considered as a viable solution to overcome this problem. Recently, Artificial Intelligence (AI) has been increasingly adopted



(a) (b)

FIGURE 1. (a) Safety helmet and (b) harness used in construction.

across various industries as an effective tool for monitoring workplace safety. Machine Learning (ML), particularly Deep Learning (DL), has been increasingly utilized for automated safety monitoring. In this study, a DL model is trained and tested to detect the presence of a safety helmet on a worker's head and the proper use of a safety harness. The trained model is based on a DEtection TRansformer (DETR) architecture.

DETR is a novel approach to object detection that leverages the power of a transformer — a neural network architecture originally developed for sequence modeling, particularly in Natural Language Processing (NLP) tasks [3]. It applies a transformer-based framework to image data, redefining object detection by treating it as a direct set prediction problem. Instead of relying on traditional methods, DETR predicts a fixed set of objects by directly generating bounding boxes and their corresponding class labels from the input image [3].

Limited research has explored the potential of using DETR as a safety monitoring tool in construction. For instance, Kim et al. [10] introduced a novel framework for detecting and classifying unsafe operations in heavy construction machinery, focusing on five types of construction equipment. Their study utilized Real-Time Detection Transformers (RT-DETR) to enhance the detection and classification of hazardous operations by upscaling low-resolution surveillance footage and generating high-resolution inputs for the RT-DETR model. Additionally, Subhi et al. [12] trained a DETR model specifically for detecting safety helmets in construction environments. However, the system exhibited a high rate of incorrect detections, including failures in identifying helmets occluded by other objects, difficulties with blurry images, and challenges in detecting small objects. These limitations hindered its ability to accurately recognize helmet components. Several factors may have contributed to these shortcomings, such as the quality of the training dataset, which may not have been sufficiently diverse to encompass the range of real-world scenarios encountered in construction fields. Additionally, the number of training epochs may have been inadequate, preventing the model from fully learning and generalizing the necessary features for accurate detection in complex environments.

This study aimed to develop a PFAS and helmet detection model capable of performing effectively and accurately under various environmental conditions and scenarios, including foggy, sandy, and humid sunny days. This was achieved by ensuring that the training dataset was diverse, incorporating a wide range of lighting, color, and blur scenarios representative of real-world conditions. Furthermore, the model was trained multiple times with varying numbers of epochs to identify the optimal configuration for enhanced detection accuracy. Beyond helmet detection, our model ensures that workers are properly equipped with both a helmet and a safety harness when working at heights, reinforcing workplace safety standards.

The model developed in this study was tested and validated under various conditions including: (1) normal RGB images, (2) grayscale images, (3) blurry images simulating foggy weather, (4) images representing dusty conditions, and (5) images captured using a drone in both real construction sites and laboratory settings. This comprehensive evaluation was conducted to assess the DETR model's performance across diverse environmental scenarios and weather conditions that may occur at construction sites. To benchmark the model's effectiveness, its results were analyzed and compared with those of YOLOv3, as reported in our previously published study [4]. YOLOv3 was selected for comparison due to its well-established accuracy and efficiency in construction safety monitoring. In our prior study, we conducted a detailed evaluation of multiple YOLO versions, including YOLOv3, v4, v5, v7, and v8, for welding helmet detection [20]. The findings indicated that YOLOv3 outperformed other versions in terms of precision and recall for detecting welding helmets. Furthermore, our results were benchmarked against two additional models developed using distinct architectures: YOLOv8 and Deformable DETR.

The main contributions of this paper are as follows:

- The introduction of a custom, diverse, and realistic dataset of safety helmets and harnesses, collected from multiple sources including drone footage, real construction sites, web images, and lab environments.

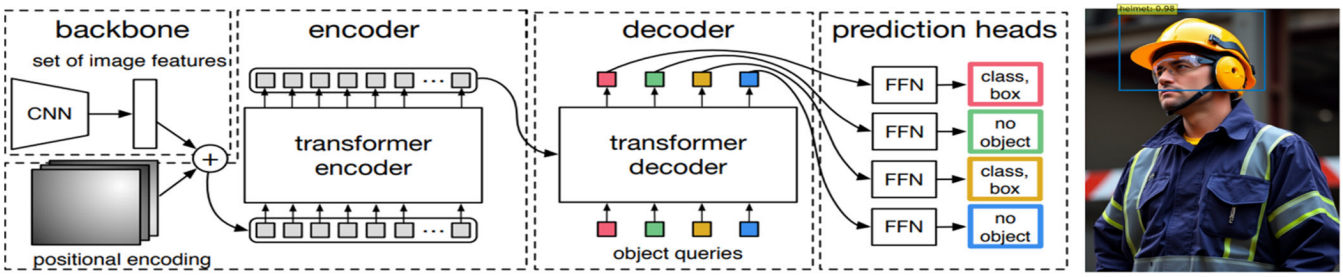


FIGURE 2. DETR Architecture consisting of backbone, transformer encoder, transformer decoder, and prediction heads [3].

This dataset better represents the complexities of real-world deployment compared to synthetic or simplified datasets.

- This research provides a comprehensive comparative analysis between DETR, Deformable DETR, YOLOv3, and YOLOv8 under identical training and evaluation conditions.
- This research conducts a systematic ablation study on critical architectural components of DETR including positional encoding types, the number of transformer layers, and the number of object queries. This provides insights into how these components affect detection performance in this domain.
- The validation of the model's effectiveness in real construction environments using drone-captured footage, addressing practical concerns of deployment, such as performance in blurry, dusty, and grayscale conditions.

The remainder of the paper is organized as follows: Section II reviews related work, highlighting previous studies that explored the application of deep learning across safety sector, including the construction industry. Section III outlines the methodology adopted in this research, which involved training a DETR-based model and comparing its performance with other widely used deep learning models, namely YOLOv3, YOLOv8, and Deformable DETR. Section IV presents the results obtained during the testing phase of the DETR model. Section V provides a performance analysis, offering a comparative evaluation of the previously mentioned models and assessing the DETR model's effectiveness in a real-world construction site scenario using drone footage. Finally, section VI concludes the paper with key findings and future research directions.

II. RELATED WORKS

Recently, numerous research studies have been published exploring the use of ML and DL models and architectures to enhance worker safety on construction sites. Various algorithms and architectures have been employed for this purpose, including YOLO, Faster R-CNN [24], [25], [26], and Single Shot Detector (SSD) [27], [28]. For instance, Yang et al. [5] proposed an algorithm for improving helmet detection in images, called YOLO-Helmet. This model extends the detection layer to better identify small objects, such as helmets, by incorporating a C-ELAN module with deformable convolution to address occlusion issues and capture richer contextual features. Thara et al. [6] utilized a YOLOv8 model to detect safety equipment worn by workers on construction sites. Similarly, Islam et al. [7] applied a YOLO v7 model to identify various types of safety gear, including helmets, goggles, jackets, gloves, and footwears worn by construction workers. The model demonstrated strong performance, enabling the rapid identification of safety equipment violations on building sites. Moreover, Shanti et al. [4] developed a novel technique using YOLOv3 to monitor workers' compliance with safety standards related to PFAS. Their approach involved a real-time detection algorithm designed to identify the key components of PFAS, including safety harness, alongside the standard safety measure of wearing a safety helmet. Jankovic et al. [8] explored the potential of lightweight, small-scale YOLOv8 architectures to detect Occupational Safety and Health Administration (OSHA) violations on construction sites, utilizing a publicly available dataset. Their results were promising, with the best-performing model achieving a precision rate of 92 %. Furthermore, Li et al. [19] employed image processing and machine learning techniques, specifically using YOLO with transfer learning, to accurately detect helmets and promote workplace safety. By fine-tuning a pre-trained YOLO model on a helmet-specific dataset, they achieved high accuracy across varying lighting conditions and perspectives, demonstrating superior robustness compared to traditional methods. Faster R-CNN has also proved its effectiveness in construction safety monitoring. For instance, Saudi et al. [9] developed an image detection model to assess workers' safety conditions based on PPE compliance, utilizing the Faster R-CNN algorithm. Similarly, Kim et al. [10] conducted research on the development of a Korean data learning model using R-CNN to identify safety protection equipment worn by workers in the field. Chen et al. [11] proposed an improved Faster R-CNN algorithm for safety helmet detection. Their experimental results indicated that, compared to the standard Faster R-CNN, the improved model achieved higher mean average precision and enabled real-time automatic detection of helmet compliance, enhancing overall workplace safety monitoring.

Moreover, Sanjeewani et al. [21] proposed a deep learning-based model for real-time identification of non-compliance with hard hat and safety vest requirements on construction sites. Their study evaluated several neural networks for edge deployment and found that the SSD MobileNet V2 model stood out for its computational efficiency, making it well-suited for real-time safety monitoring applications. Arfan et al. [22] employed the SSD algorithm to detect and monitor PPE usage across more than 132 different construction site scenarios. Furthermore, Zhan et al. [23] developed an improved SSD model for helmet detection. Their results demonstrated that the model successfully identified helmets in over 94.5% of the test dataset.

III. METHODOLOGY

A. DETR VS YOLOV

DETR is an object detection architecture that integrates CNNs with transformers to model the relationships between objects in an image [3]. The DETR framework typically comprises four main components: the backbone, the transformer encoder, the transformer decoder, and the prediction heads-also known as prediction feed-forward networks (FFNs)-as illustrated in Figure 2.

The CNN backbone processes the input image to generate a feature map, which includes positional encodings to help the model capture spatial information. This feature map is then processed by the transformer encoder, which captures the global context across the entire image. Simultaneously, object queries are fed into an input and streamed through the transformer decoder, allowing the model to focus on different regions of the feature map. Finally, the prediction heads generate a bounding box and an object class prediction for each query. DETR employs a bipartite matching algorithm to associate ground-truth objects (y_k) with predicted objects $\left(\hat{y}_{\sigma(k)}\right)$ by minimizing a predefined matching cost, as expressed by the following equation:

$$L_m\left(y_k, \hat{y}_{\sigma(\mathrm{k})}\right) = -1_{\{c_k \neq\}} \hat{p}_{\sigma(\mathrm{k})}\left(c_k\right) + 1_{\{c_k \neq\}} L_{\mathrm{bbox}}\left(b_k, \hat{b}_{\sigma(\mathrm{k})}\right) \quad (1)$$

where c_k represents the ground-truth class label, b_k denotes the ground-truth bounding box, $\hat{p}_{\sigma(\mathrm{k})}$ is the predicted class probability, and $\hat{b}_{\sigma(\mathrm{k})}$ is the predicted bounding box. The total matching cost is minimized using the following equation:

$$\hat{\sigma} = \arg\min_{\sigma \epsilon N} \sum_{k=1}^{N} L_m\left(y_k, \hat{y}_{\sigma(\mathrm{k})}\right). \quad (2)$$

Once the matching process is completed, DETR calculates the Hungarian loss to optimize both classification and bounding box predictions:

$$L_H\left(y, \hat{y}\right) = \sum_{k=1}^{N}\left[-\log \hat{p}_{\sigma(\mathrm{k})}\left(c_k\right) + 1_{\{c_k \neq\}} L_{\mathrm{box}}\left(b_k, \hat{b}_{\sigma(\mathrm{k})}\right)\right]. \quad (3)$$

The bounding box loss (L_{BBox}) combines generalized IoU (GIOU) loss and L1 loss as:

$$L_{\mathrm{bbox}} = \lambda_i L_{\mathrm{Giou}}\left(b_k, \hat{b}_{\sigma(\mathrm{k})}\right) + \lambda_{11}\left|b_k - \hat{b}_{\sigma(\mathrm{k})}\right|_1 \quad (4)$$

where λ_i and λ_{l1} are weighing hyperparameters.

Finally, Hungarian matching assigns predicted objects to ground-truth objects, upon which a loss function is computed based on the matching results. The Hungarian loss function consists of two main components, i.e., classification loss and bounding box loss. For each predicted object i, matched with a ground-truth object j, the total loss L is defined as follows:

$$L\left(\mathrm{i}, \mathrm{j}\right) = w_c\left[-\log\left(\hat{p}_{\sigma(k)}\left(c_k\right)\right)\right] + w_b\left|b_{\mathrm{k}} - \hat{b}_{\sigma(\mathrm{k})}\right|_1 + w_g L_{\mathrm{GioU}}\left(b_k, \hat{b}_{\sigma(k)}\right) \quad (5)$$

where, w_c, w_b, and w_g are weights applied to balance the different loss terms.

In Summary, Equation (1) defines the matching loss, which combines classification and bounding box components to assess the quality of predicted assignments. Equation (2) refines this process by applying the Hungarian algorithm determine the optimal assignment by minimizing the total matching cost. Once matching is established, Equation (3) introduces the Hungarian loss, integrating classification loss and bounding box loss for model training. The bounding box loss, detailed in Equation (4), incorporates Generalized Intersection over Union (GIoU) and L1 losses to enhance localization accuracy. Finally, Equation (5) generalizes this framework by representing the total loss as a weighted combination of classification and bounding box losses. This structured approach ensures a consistent methodology for aligning predictions with ground-truth data while optimizing object detection performance.

YOLO and DETR differ significantly in their approaches to object detection. YOLO models are convolutional-based, single-stage detectors that directly predict bounding boxes and class probabilities from the input image in a dense, end-to-end manner. Earlier versions (e.g., YOLOv3, YOLOv4) rely on anchor boxes and divide the image into a grid, predicting objects relative to each cell, followed by Non-Maximum Suppression (NMS) to remove redundant detections. More recent versions, such as YOLOv8, adopt an anchor-free design with decoupled heads, offering enhanced flexibility and performance, while still maintaining the core principle of fast and accurate detection. YOLO's loss function typically includes objectness, classification, and localization components, tuned to balance precision and speed. In contrast, DETR introduces a transformer-based architecture that treats object detection as a direct set prediction problem. It eliminates the need for anchor boxes, grid partitioning, and NMS by employing a fixed number of learned object queries and leveraging bipartite Hungarian Matching to associate predictions with ground truth. This allows DETR to model complex global relationships in the image. Although

TABLE 1. Comparison between YOLO and DETR architectures.

Feature	YOLO	DETR
Architecture	CNN-based (evolved to include modern modules in later versions)	Transformer-based end-to-end detection
Object Detection Approach	Grid-based prediction (anchor-based/anchor-free)	Set-based prediction using learned object queries
Anchor Boxes	Used in earlier versions (YOLOv3), anchor-free in YOLOv8	Anchor-free
Post-Processing	Requires Non-Maximum Suppression (NMS)	No NMS; uses Hungarian Matching
Multi-Scale Predictions	Yes (feature pyramid or PANet in newer versions)	Initially no, added in Deformable DETR
Best for	Real-time applications, mobile/edge deployment	High-precision detection, complex scenes with occlusion
Model Complexity	Lightweight to moderate (depending on version)	Architecturally more complex due to attention mechanism

DETR tends to have higher inference latency compared to YOLO due to its attention mechanism and decoder stages; it offers advantages in terms of simplicity in design and performance on complex scenes with overlapping objects. Table 1 highlights a comparison of their key architectural differences.

B. RESEARCH METHODOLOGY

The proposed research methodology is structured into several key stages. The process begins with the acquisition of both training and testing datasets. Once the dataset is collected, the model undergoes training using the acquired data. Following the training phase, the model is evaluated to assess its performance and determine whether it meets the desired criteria for acceptability. Upon achieving satisfactory results, additional evaluation tests are conducted to ensure the model's reliability and robustness under various conditions, including grayscale images, foggy weather, dusty environments, and images captured using a drone. Subsequently, a comprehensive comparison is made between the results obtained from the DETR model and those from a YOLOv3 model previously used for the same objective. A flowchart illustrating the research design and methodology is presented in Figure 3.

C. MODEL'S DATASET

The dataset used to evaluate the DETR model's performance was compiled from a diverse range of sources, including real construction sites, publicly available web images, industry contributions, controlled lab environments, and the open-source 'Safety Helmet Detection' dataset from Kaggle to ensure both realism and diversity [32]. The dataset was divided into two subsets, i.e., 70% for training and 30% for testing. The images featured helmets and harnesses of various types, angles, and colors, ensuring a diverse dataset capable of detecting helmets across different construction site scenarios. Figure 4 presents examples of the images used for training the model.

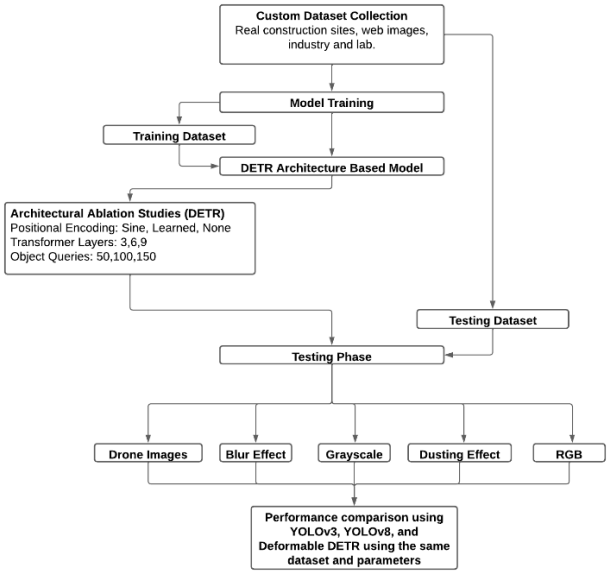


FIGURE 3. Flowchart of research methodology.

D. PSEUDO CODE OF THE RESEARCH METHOD

To clearly illustrate the proposed DETR-based object detection framework, Figure 5 shows a pseudo code that outlines the key steps of the training and inference pipeline. This abstraction highlights the core structure of our approach without relying on implementation-specific details.

IV. TRAINING AND TESTING OF DETR

A. TRAINING

The DETR model was trained and tested using the obtained dataset. In this study, Google Colab was utilized for model training. Google Colab is a cloud-based platform that enables users to write and execute Python code, particularly for machine learning tasks, while providing access to powerful computing resources. Its availability of powerful GPUs significantly accelerated the model training process. For this study, an Intel Xeon CPU with 16 GB RAM and an NVIDIA



FIGURE 4. Sample images used in creating the diverse training dataset (a) Clear weather, sunny, high visibility, simple background (b) Clear weather, daylight, scaffold scene (c) Overcast lighting, moderate background, clear foreground (d) Clear weather, simple background, ladder work (e) Overcast, rocky terrain, moderate visibility (f) Low lighting, underground, high clutter (g) Harsh lighting, high contrast, (h) Sunny, outdoor, multiple workers [14], [15], [16], [17], [32].

Algorithm 1: DETR

Input: Training dataset (images and annotations)

Output: Trained DETR model for helmet and harness detection

Begin

1. Initialize DETR model with:
   - Backbone CNN
   - Transformer encoder and decoder
   - Set of learnable object queries
   - Positional encoding
2. For each training epoch:
   a. For each image in training dataset:
      - Extract feature map using the CNN backbone
      - Add positional encoding to the feature map
      - Encode the features using the Transformer encoder
      - Decode features using Transformer decoder and object queries
      - Predict bounding boxes and class labels
   b. Compute loss using:
      - Hungarian Matching for bipartite matching between predictions and ground truth
      - Loss functions: classification loss + box regression loss + cardinality loss
   c. Backpropagate and update model weights
3. Validate model using standard metrics (e.g., mAP, IoU)
4. Test the model on unseen data
5. Perform architectural ablation studies by varying:
   - Type of positional encoding
   - Number of transformer layers
   - Number of object queries
6. Compare results with other models (YOLOv3, YOLOv8, Deformable DETR)

return trained DETR model

FIGURE 5. Pseudo code of the research method.

Tesla GPU (T4) were used to train the model. The training process was optimized based on key hyper-parameters, including batch size, number of epochs, number of object queries, and learning rate. Table 2 presents a summary of the values assigned to each hyper-parameter. These values were determined through multiple trial-and-error testing to achieve the best performance. Furthermore, the AdamW optimizer was employed in training the model. AdamW is an optimization algorithm that decouples weight decay from the gradient update process, enhancing regularization and improving generalization in deep learning models [31]. Figure 5 illustrates the loss versus epoch graph from the training process, demonstrating a minimal final loss value of approximately 0.0329. The figure also compares the performance of the trained DETR model with the previously trained YOLOv3 model. Despite being trained for fewer epochs, DETR exhibited superior performance throughout the training process. Specifically, while YOLOv3 required 130 epochs, DETR achieved better results within just 100 epochs.

Moreover, Deformable DETR and YOLOv8 models were also trained on the same dataset and parameters for comparison. The training losses of the models are illustrated in Figure 6. Among all four models, the DETR model achieved the lowest training loss, indicating superior performance during training. The figure also presents the results of YOLOv3 trained for 100 epochs, in contrast to its previously determined optimal performance at 130 epochs at our previous study. This adjustment was made to ensure a fair and unbiased comparison of all models under consistent hyperparameter settings.

Additionally, the DETR model was evaluated using mean Average Precision (mAP) across multiple Intersection over Union (IoU) thresholds, ranging from 0.5 to 0.95 with a step size of 0.05, in order to provide a more robust and comprehensive assessment of its performance. The mAP

TABLE 2. Model training parameters.

Batch Size	Epochs	Number of Object Queries	Learning Rate
16	100	100	0.001

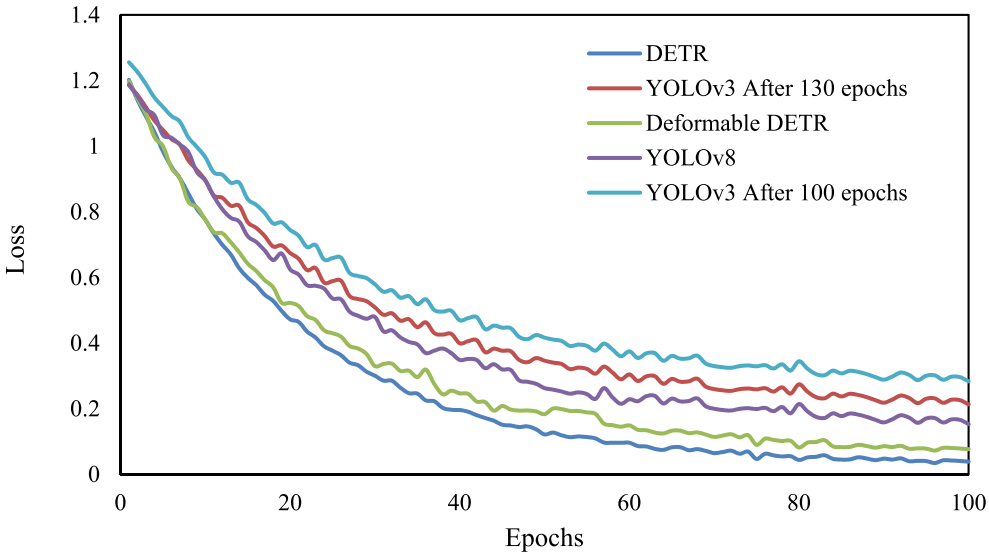


FIGURE 6. Comparison of training performance of DETR, Deformable DETR, YOLOv8, and YOLOv3.

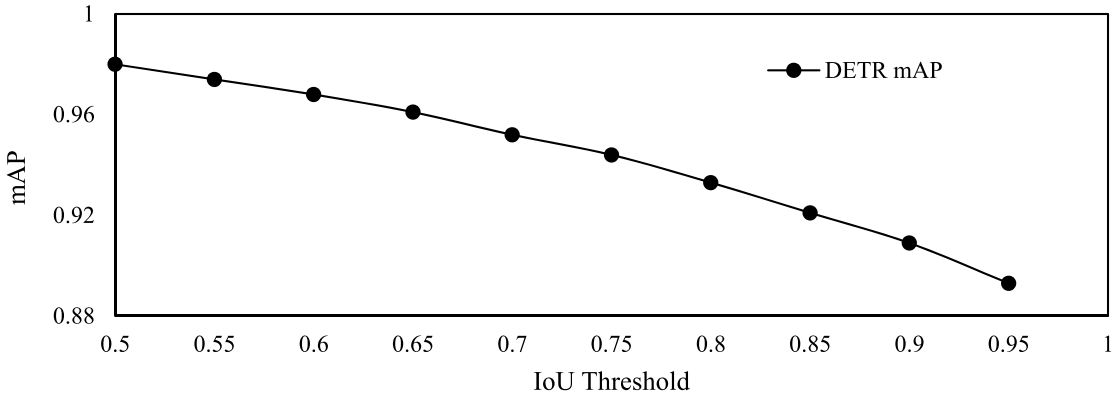


FIGURE 7. Mean Average Precision (mAP) of the DETR model across multiple IoU thresholds.

values across these thresholds were used to generate a precision-IoU curve, as shown in Figure 7.

The results demonstrate that DETR achieves high precision at the lower IoU thresholds (e.g., mAP@0.5 = 0.982), reflecting its strong object detection capability. As the IoU threshold increases, the precision gradually decreases, which is expected due to the stricter requirement for accurate localization. However, the performance remains relatively stable and acceptable even at higher IoU thresholds (e.g., mAP@0.95 = 0.63), indicating that DETR maintains reliable detection quality across varying levels of localization strictness.

After completing the training phase, the model was evaluated through a structured testing process divided into four sections. First, the model was tested using standard RGB images. Next, these images were converted to grayscales to assess the model's performance in environments with "no color" information (e.g., Figure 8(b) and (d)). Subsequently, the images were blurred to simulate foggy and humid conditions (e.g., Figure 9(d)). Finally, the dataset was modified to depict dusty (sandy) conditions (e.g., Figure 8(b)). These testing stages were conducted to evaluate the model's reliability and robustness across diverse conditions encountered on construction sites.

B. TESTING

The DETR model was evaluated using both RGB and grayscale images to assess its performance across varying lighting and color conditions. Figure 8 presents a comparison of the DETR model's detection accuracy on the same images in both RGB and grayscale formats. The results indicate that the model was highly effective in detecting safety helmets



FIGURE 8. RGB and Grayscale (a) RGB image sample 1 (b) Grayscale image sample 1 (c) RGB image sample 2 (d) Grayscale image sample 2.



FIGURE 9. Dusting and Blurry Effects sample results (a) RGB image (b) Dusting effect image (c) RGB image (d) Blurring effect image.

in both scenarios. The model identified all helmets and harnesses present in the images, demonstrating smooth and accurate detection even in cases where multiple helmets were visible. Notably, the model achieved an almost perfect confidence score (1.00) in both RGB and grayscale images. Furthermore, the DETR model's performance was tested under various weather conditions. The testing dataset was modified to simulate foggy, humid, and dusty environments.



(a) (b)

FIGURE 10. Sample test images from kaggle dataset showcasing different scenarios of helmet detection: (a) individuals with and without wearing helmet and (b) various sizes of helmets.

Figure 9 shows examples of the detection results under these conditions. As illustrated, the model maintained high accuracy and confidence in its detections, despite the applied effects. Similar to previous evaluations, the model consistently achieved near-perfect confidence scores (1.00) across different weather conditions, underscoring its robustness and reliability in diverse construction site scenarios.

To demonstrate the effectiveness of our model, we included a series of detection images showcasing various challenging scenarios. Figure 10 highlights the model's ability to accurately distinguish between individuals wearing helmets and those without, even in mixed environments where both cases are present simultaneously. Furthermore, the model successfully detects smaller-sized helmets and is capable of identifying multiple helmets within a single frame. These results underline the robustness of the model in complex and realistic settings, emphasizing its potential for real-world deployment in safety monitoring applications. These images were taken from real construction sites from a Kaggle data set under the name of "Safety Helmet Detection" [32]. The model showed an average of 99% confidence in detecting the helmets in these images.

V. DISCUSSION OF RESULTS

After evaluating the model, several key performance metrics obtained were analyzed. This study considered multiple deep learning evaluation criteria, including: (1) Precision, (2) Recall, (3) F1 Score, and (4) Detection Confidence. These metrics were computed to assess the model's reliability and consistency in object detection across the four different scenarios discussed in the previous section. These evaluation parameters are closely interrelated. Precision measures the proportion of correctly identified positive instances among all predicted positives, with higher precision indicating fewer False Positive (FP) predictions. Conversely, recall measures the proportion of actual positive instances correctly identified by the model, where a high recall signifies the model's ability to detect most relevant instances within the dataset. The F1-score, which is the harmonic mean of precision and recall, provides a balanced metric that accounts for both false positives and false negatives [18]. The formulas used to compute these metrics are as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \tag{6}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \tag{7}$$

$$\text{F1Score} = \frac{2\,(\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}. \tag{8}$$

where the True Positives (TP) represent the number of correctly detected objects (e.g., helmets or harnesses), while False Positives (FP) denote the number of mistakenly detected objects. False Negatives (FN) indicate the number of undetected objects. Table 3 summarizes the overall TP, FP, and FN values for all cases, along with the corresponding precision, recall, and F1 score for helmet detection. Similarly, Table 4 provides a summary of these values for safety harness detection. The obtained precision, recall, and the F1 score values were consistently high, demonstrating the model's effectiveness. As shown in the results, the model achieved a precision above 99% in almost all cases, indicating its strong ability to accurately detect the presence of helmets and harnesses. Additionally, the recall values for RGB, grayscale, and blurry images remained high, with a slight

decrease observed in the dusty weather scenario. The high recall values suggest that the model effectively identified most, if not all, instances of the target objects, even under varying visual conditions. These results demonstrate the robustness of the model in accurately identifying objects, despite potential challenges posed by varying image quality and color conditions. Additionally, the F1 score, which balances precision and recall, further validates the model's effectiveness in accurately detecting safety gear while minimizing false positives and false negatives. The consistently high performance across diverse scenarios underscores the model's reliability and generalizability in real-world construction environments. As shown in Tables 3 and 4, the model achieved higher performance metrics for helmet detection compared to harness detection. This disparity can be attributed to the helmet's greater visibility when worn and its simpler, more distinguishable shape, making it easier to detect. In contrast, harnesses, which are worn over the body, often blend with the workers' clothing, increasing the likelihood of misdetection.

TABLE 3. DETR evaluation metrics for safety helmets.

	RGB	Grayscale	Blurry	Dusty
TP	243	225	231	189
FP	1	2	4	2
FN	7	14	17	39
Precision	0.996	0.992	0.983	0.99
Recall	0.972	0.942	0.932	0.83
F1 Score	0.984	0.966	0.957	0.903

TABLE 4. DETR evaluation metrics for safety harnesses.

	RGB	Grayscale	Blurry	Dusty
TP	240	228	236	228
FP	0	8	12	8
FN	16	24	20	32
Precision	1.000	0.966	0.952	0.966
Recall	0.938	0.905	0.922	0.877
F1 Score	0.968	0.934	0.937	0.919

Moreover, a detailed study was conducted to evaluate the impact of key architectural hyperparameters including positional encoding, the number of transformer layers, and the number of object queries, on the overall performance of the model. The baseline DETR model, used throughout the study, was configured with sine positional encoding, 6 transformer layers, and 100 object queries. To assess the influence of positional encoding, the model was also trained using no positional encoding and learnable positional encoding. Similarly, the effect of varying the number of transformer layers was explored by testing configurations with 3, 6, and 9 layers. Lastly, to understand the role of object queries, experiments were conducted using 50, 100, and 150 queries. Table 5 shows a summary of the results obtained.

It is evident that removing positional encoding significantly degrades the model's performance, confirming its critical role in spatial awareness for detection. Replacing sine-based encoding with learned positional embeddings results in slight performance gains in recall, but marginally lower precision.

Reducing transformer layers from 6 to 3 causes a noticeable drop in all metrics, indicating insufficient depth for effective feature modelling. Increasing the number of layers to 9 slightly improves precision but has minimal impact on F1 Score, suggesting diminishing returns beyond 6 layers.

As for object queries, using fewer (50) reduces recall and F1, likely due to missed detections. Increasing to 150 queries marginally improves all metrics, showing the benefit of a higher query capacity for complex scenes.

TABLE 5. Architectural hyperparameters comparison.

Configuration	Precision	Recall	F1 Score
Baseline DETR	**0.981**	**0.915**	**0.947**
No Positional Encoding	0.951	0.882	0.915
Learnable Positional Encoding	0.9602	0.895	0.9265
3 Transformer Layers	0.948	0.868	0.905
9 Transformer Layers	0.956	0.886	0.918
50 Object Queries	0.942	0.859	0.898
150 Object Queries	0.954	0.88	0.915

Another key metric used to evaluate the model's performance was the confidence rate. This value, displayed alongside detected objects, such as "helmet" or "harness," represents the model's certainty in its classification. The DETR model of this study consistently demonstrated high detection confidence, exceeding 93% across all test cases. Table 6 provides a summary of the average confidence percentage for each case.

TABLE 6. Confidence (%) in detections.

	RGB	Grayscale	Blurry	Dusty
Harness	95.52	93.43	93.55	93.33
Helmet	97.89	96.85	92.65	94.66

A. COMPARISON WITH OTHER MODELS

Following the analysis of the trained model's results, a comparative evaluation was carried out against a previously developed YOLOv3 model [4], which was designed for the same objective. Additionally, the performance of the model was compared to two other models, one trained using the YOLOv8 algorithm and the other using the Deformable DETR architecture, both trained on the same dataset and parameters. This comparison aimed to benchmark the results against the base DETR model. Table 7 and 8 show the results obtained after testing these models for the safety helmet and Tables 9 and 10 for the harness.

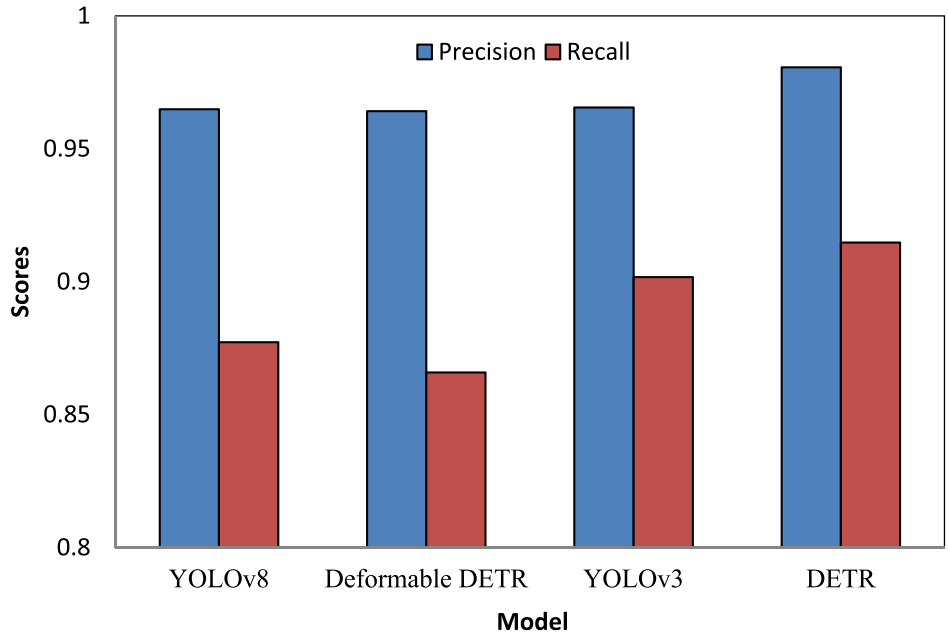
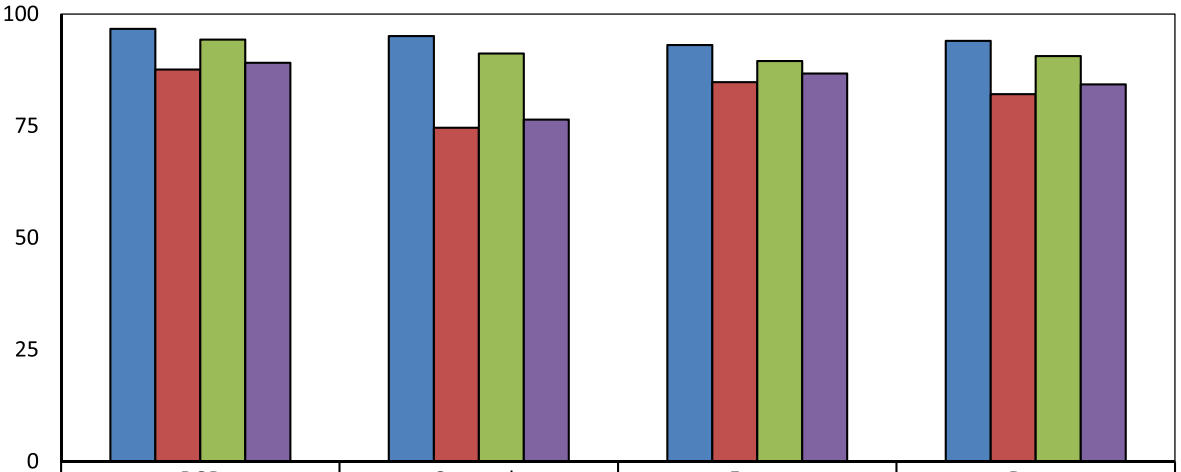


FIGURE 11. Model's precision and recall comparison.



	RGB	Grayscale	Foggy	Dusty
DETR	96.7	95.1	93.1	94
YOLOv3	87.6	74.6	84.8	82.1
Deformable	94.3	91.2	89.5	90.6
YOLOv8	89.1	76.4	86.7	84.3

FIGURE 12. Confidence Comparison for all models.

The DETR model outperformed all models in nearly all scenarios, except for the dusting effect when compared to YOLOv3. Additionally, it required less training time and epochs while achieving higher accuracy throughout the training process. Figure 11 shows a comparison of precision, recall, and F1 score for all models, displaying the average metric values for each model.

A significant disparity between the models was observed in the confidence scores of their detections. The DETR model consistently exhibited higher confidence values compared to the other models. A higher confidence score indicates a more reliable and trustworthy prediction. In all test cases, the DETR model achieved confidence rates exceeding 93%. Figure 12 illustrates the comparison of the average confidence rates for all models. The superior confidence levels of DETR can be attributed to its Transformer architecture, which effectively captures global context and spatial relationships through self-attention mechanisms. This design

TABLE 7. Deformable DETR evaluation metrics for safety helmet.

	RGB	Grayscale	Blurry	Dusty
TP	220	195	200	150
FP	5	6	9	6
FN	20	28	32	60
Precision	0.978	0.97	0.957	0.962
Recall	0.917	0.874	0.862	0.714
F1 Score	0.946	0.92	0.907	0.823

TABLE 8. YOLOv8 evaluation metrics for safety helmet.

	RGB	Grayscale	Blurry	Dusty
TP	235	210	220	170
FP	3	5	6	4
FN	12	20	25	50
Precision	0.987	0.976	0.973	0.977
Recall	0.951	0.913	0.898	0.773
F1 Score	0.969	0.943	0.933	0.862

TABLE 9. Deformable DETR evaluation metrics for safety harnesses.

	RGB	Grayscale	Blurry	Dusty
TP	235	220	230	215
FP	2	10	14	10
FN	20	28	25	38
Precision	0.992	0.956	0.942	0.956
Recall	0.921	0.887	0.902	0.85
F1 Score	0.956	0.92	0.921	0.9

TABLE 10. YOLOv8 evaluation metrics for safety harnesses.

	RGB	Grayscale	Blurry	Dusty
TP	228	210	220	205
FP	4	12	16	12
FN	24	32	30	42
Precision	0.983	0.946	0.932	0.945
Recall	0.905	0.868	0.88	0.83
F1 Score	0.942	0.904	0.905	0.883

enables precise and reliable predictions while eliminating overlapping and duplicating detections, resulting in cleaner outputs and better performance in complex or cluttered scenarios. Furthermore, Figure 13 provides a side-by-side comparison of the confidence rates for all models on the same test image. As shown, Figure 13 (a), which utilizes the DETR architecture, demonstrates higher confidence rate than Figure 13 (b),(c) and (d).

The DETR model achieved better accuracy and robustness compared to YOLOv3 in our helmet and harness detection task. Specifically, DETR outperformed YOLOv3 in precision and recall metrics, with clearer detection of safety harnesses even under partial occlusion and low-contrast grayscale images. YOLOv3 struggled with false negatives in crowded construction scenes and often failed to distinguish between visually similar safety gear. In contrast, DETR's transformer-based architecture captured long-range dependencies, allowing it to localize objects more precisely in challenging scenes.



FIGURE 13. Comparison between different architectures in terms of confidence: (a) DETR, (b) YOLOv3, (c) YOLOv8, (d) Deformable DETR.

While YOLOv3 offered faster inference speeds, DETR provided more reliable detection results, particularly in real drone-captured images from construction sites, where object scale and positioning varied significantly. Moreover, DETR's performance remained stable across different architectural ablation settings, indicating its adaptability.

The extended comparison with YOLOv8 and Deformable DETR also reinforced DETR's robustness. While YOLOv8 offered high speed and good accuracy, it showed sensitivity to background noise. Deformable DETR improved speed over standard DETR but required more memory. Overall, DETR demonstrated the best trade-off between accuracy, consistency, and architectural simplicity in this task.

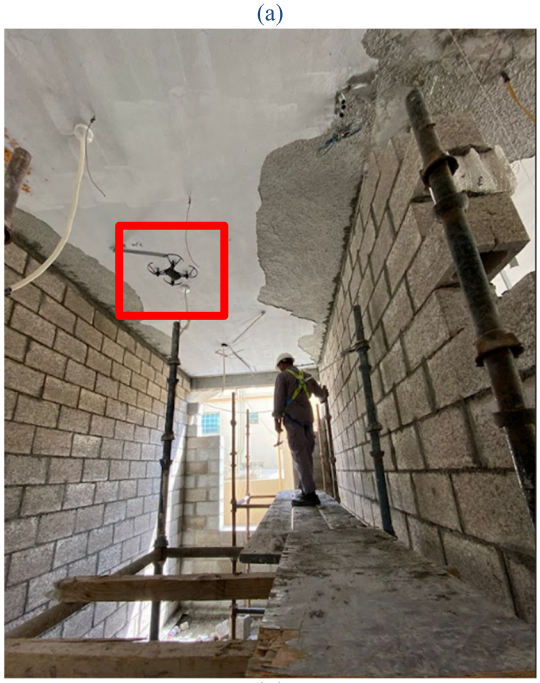
B. DRONE IMAGES

To further assess the performance of the trained DETR model, we utilized a Tello drone (shown in Figure 14) to capture images of real construction workers and conduct object detection. Figure 14 (b) illustrates the drone hovering around the site, capturing images while workers perform their tasks.

The results obtained were then compared with those from the previously tested YOLOv3 model [4]. This experiment was crucial in evaluating the model's reliability in real-world scenarios. Figure 15 presents a sample of the detection results alongside a comparison with the YOLOv3 model. The DETR model demonstrated high accuracy in detecting both helmets and harnesses, with significantly improved confidence scores compared to the YOLOv3 model. Specifically, the DETR model achieved a confidence score of 0.99 for both helmet and harness detections, whereas the YOLOv3 model recorded confidence scores of 0.89 for harness detection and 0.75 for helmet detection.



(a)

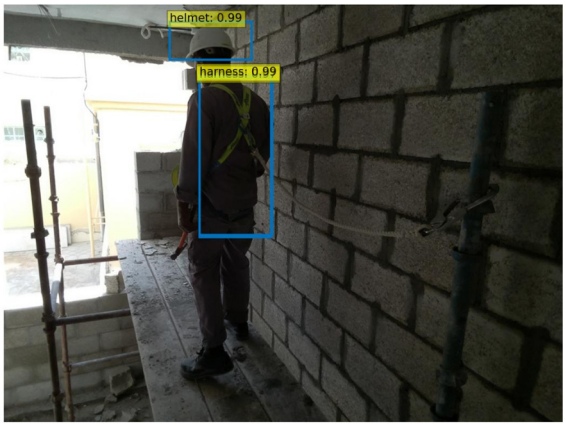


(b)

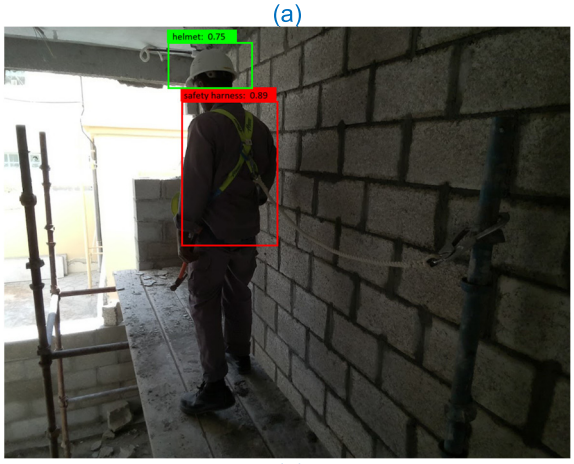
FIGURE 14. (a) Tello drone and (b) Tello drone hovering around workers capturing images.

C. DEPLOYMENT CONSIDERATIONS

To evaluate the feasibility of deploying the DETR model in practical environments, we assessed its performance on a Google Colab Tesla T4 GPU setup. The model achieved an average inference speed of 15–18 frames per second (FPS),with GPU utilization averaging 80%,VRAM consumption around 1800MB, and CPU usage approximately 30%. These metrics suggest that DETR can be effectively used for real-time monitoring tasks in construction settings, especially when accuracy and robustness are critical. Additionally, DETR's end-to-end transformer-based architecture reduces the need for manual post-processing, simplifying maintenance and improving scalability. Table 11 shows a comparison between the different architectures. Although DETR is not the fastest in terms of inference speed, it shows an excellent balance between computational efficiency and robustness. Its high accuracy and resilience in complex construction scenes make it particularly well-suited for detecting the availability of safety equipment such as helmets and harnesses in real-time monitoring applications.



(a)



(b)

FIGURE 15. Comparison between DETR and YOLOv3 architecture using drone images (a) DETR (b) YOLOv3.

TABLE 11. Performance comparison of the models for helmet and harness detection in terms of speed, resource usage, and robustness.

Feature	YOLOv3	YOLOv8	Deformable DETR	DETR
Inference Speed (FPS)	24–26	21–23	13–15	15–18
GPU Utilization	~60%	~70%	~85%	~80%
VRAM Usage	~1200MB	~1400MB	~2200MB	~1800MB
CPU Usage	~25%	~28%	~35%	~30%
Robustness in Complex Scenes	Moderate	Moderate	High	High

VI. CONCLUSION

Safety helmets and safety harnesses play a crucial role in protecting workers on construction sites. In this research, a safety monitoring detection model was developed using DETR architecture. The model was trained on a custom dataset consisting of images of helmets and harnesses. To assess its reliability, the model underwent a comprehensive testing phase across five different scenarios: (1) RGB images (2) Grayscale images (3) Blurring effect (simulating foggy

conditions) (4) Dusting effect (simulating sandy conditions), and (5) Drone-captured images. These tests ensured the model's ability to perform accurately under varying color and weather conditions. In this research, four key evaluation metrics were used to measure the model's performance: precision, recall, F1 score, and confidence. The model achieved high scores across all metrics, with precision values of 0.99 for helmets and 0.971 for harnesses, recall values of 0.92 for helmets and 0.91 for harnesses, F1 scores of 0.95 for helmets and 0.94 for harnesses, and confidence levels of 95.5% for helmets and 94 % for harnesses. A comparative analysis was conducted against three models: the previously designed YOLOv3, and two newly trained models using YOLOv8 and Deformable DETR architectures. The baseline DETR model outperformed all models in nearly all evaluation metrics and scenarios, The strong performance of the DETR model demonstrates its potential as an effective safety monitoring tool in construction environments. For future research, DETR could be explored as a monitoring tool for detecting all PPE items on construction sites.

APPENDIX

A unique dataset was generated especially for the purpose of this research. The dataset produced for this research is available upon request to the corresponding author.

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